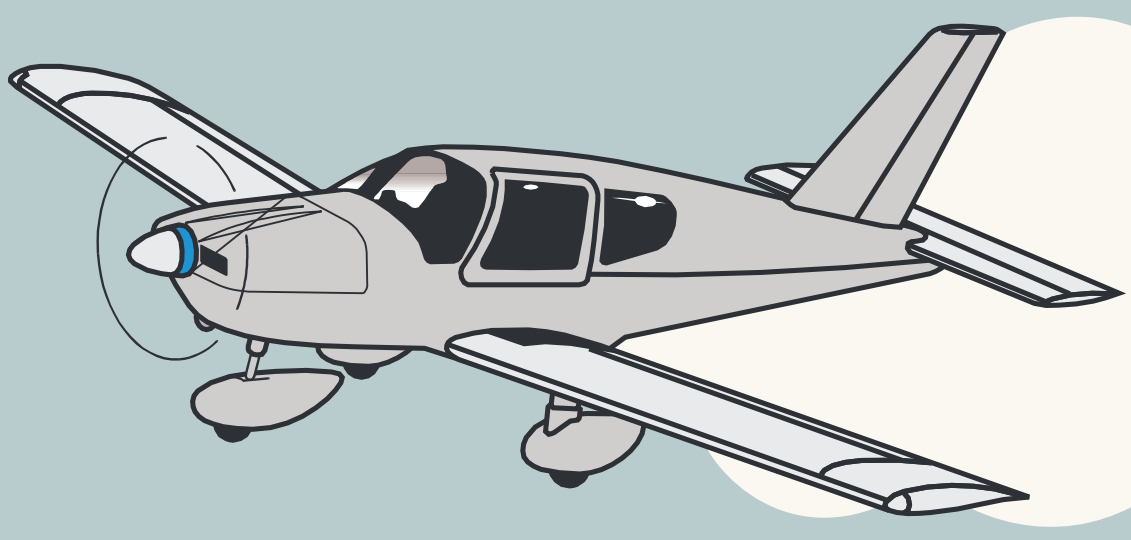




# SKYDIVING

Jannis Sacht, Finn Nannt,  
Mehmet Ünlü and  
Robert Bruns  
Group 1

## How dangerous is skydiving? What factors influence the risk of accidents? How can safety be optimized through training, equipment, and decision-making?



### Motivation

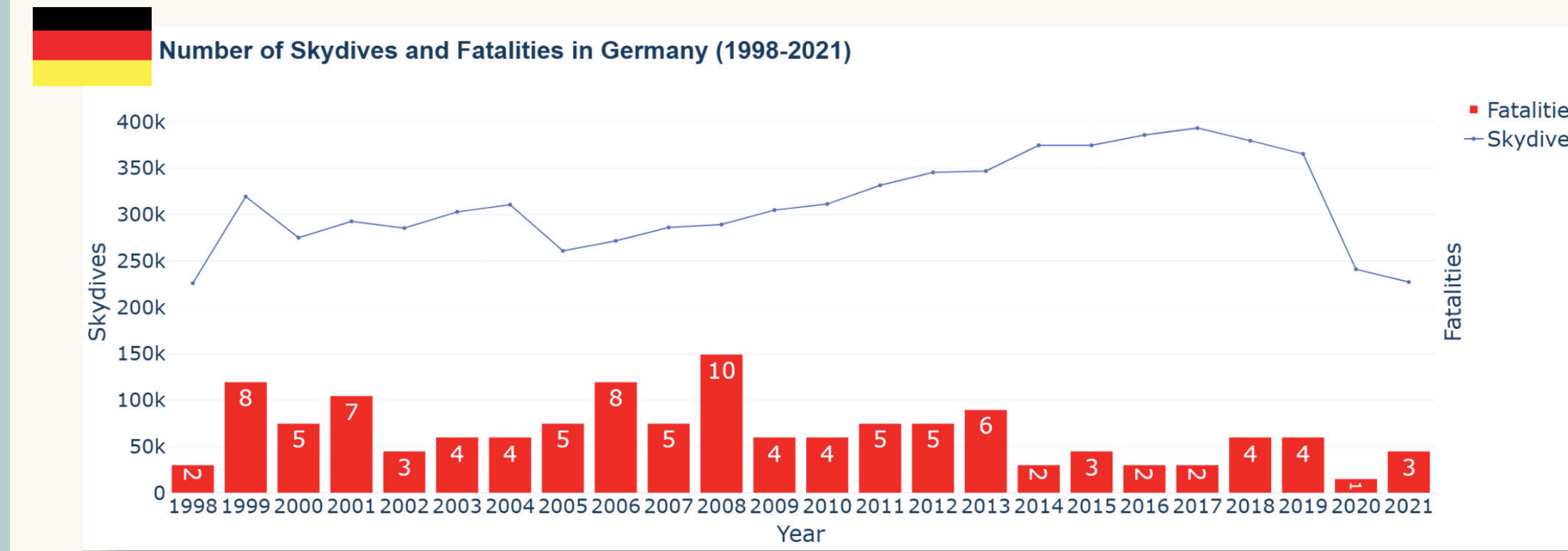
The analysis of data on skydiving incidents does not only provide a cold hard look at the chances of survival for anyone contemplating to try the sport themselves, but also provides valuable insights for those already in it, to learn from past mistakes and make their hobby the safest it can be. Since one member of our group regularly does skydiving and is active in the community, our analysis might prove useful in the future.

### Data Collection and Method

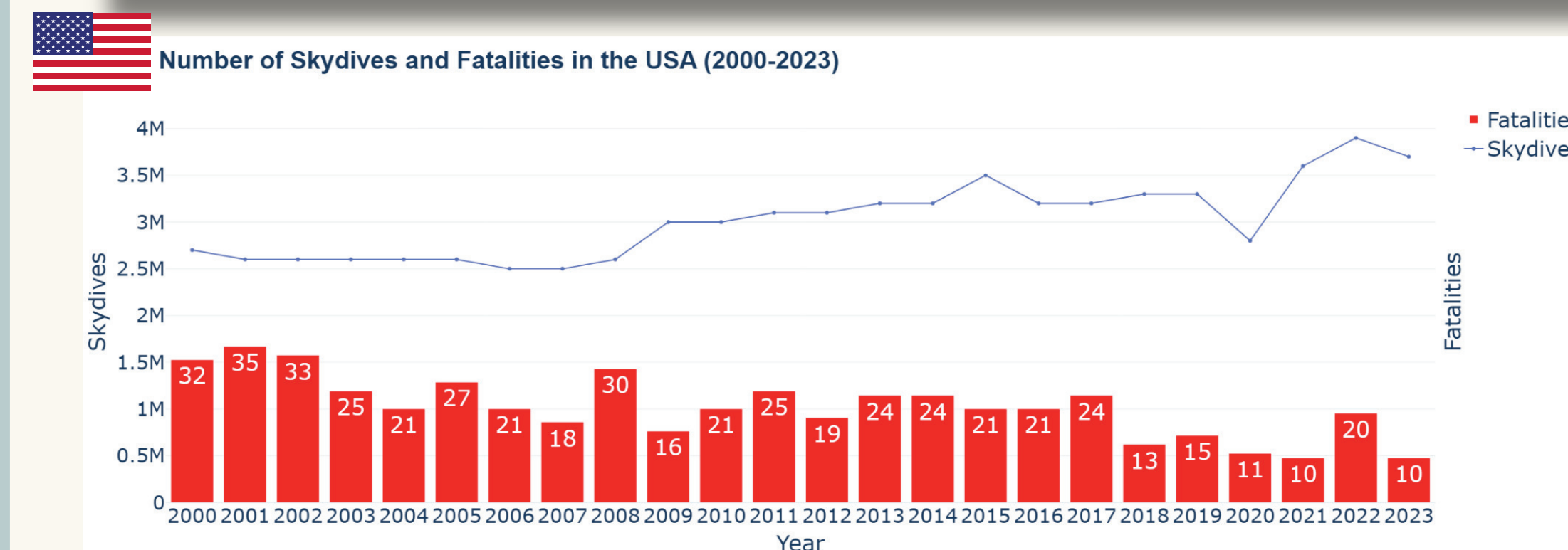
- We used the Base Fatality List (BFL), an international collection of all known fatalities as a result of BASE jumping.
- Furthermore, we looked at data from the United States Parachute Association (USPA) for skydiving accidents in the US.
- For both data sources we had to write Python scrapers, because the sites did not offer a CSV file for the data.
- For Germany, only PDF files with summary statistics and select accident reports were available through the DFV. These needed to be manually scraped.
- Overall, available data was limited. We also used additional data from Statista.



### How did the overall safety develop?



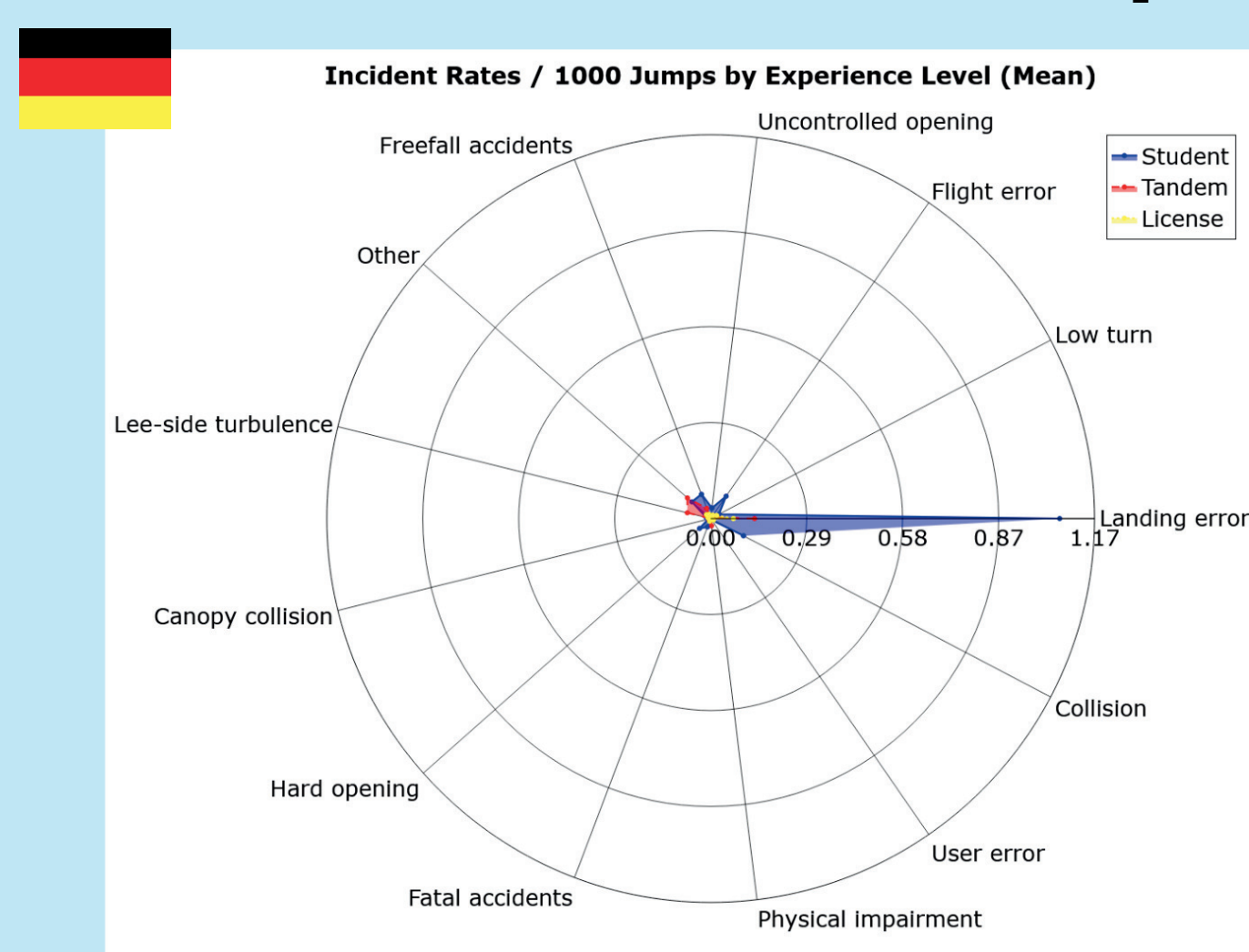
- Overall, one in 68.843 jumps ended in a fatality.
- For 2012-2021 this dropped to one in 106.121.



- In the US, overall one in 134.958 jumps ended fatal.
- For 2012 to 2021 it is one in 177.359.

This shows, that skydiving became safer.

### How does the Experience Level factor in?

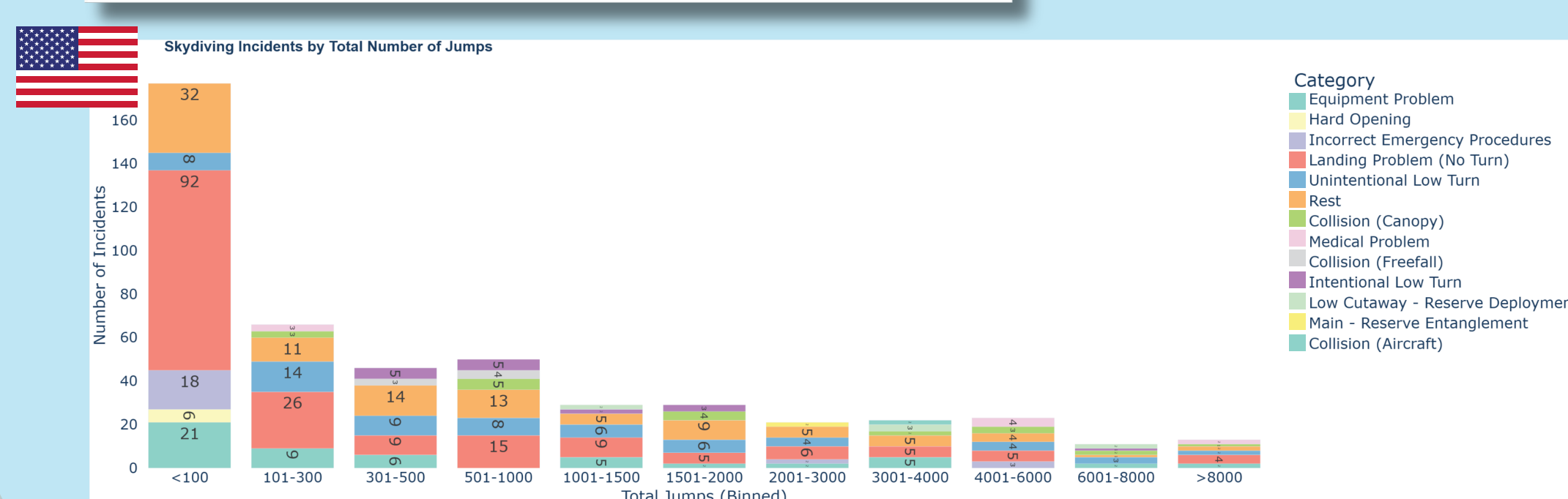


#### Germany

- Inexperienced skydivers are more prone to landing accidents.
- This is also the most prominent accident category across all levels, but the students do stand out.

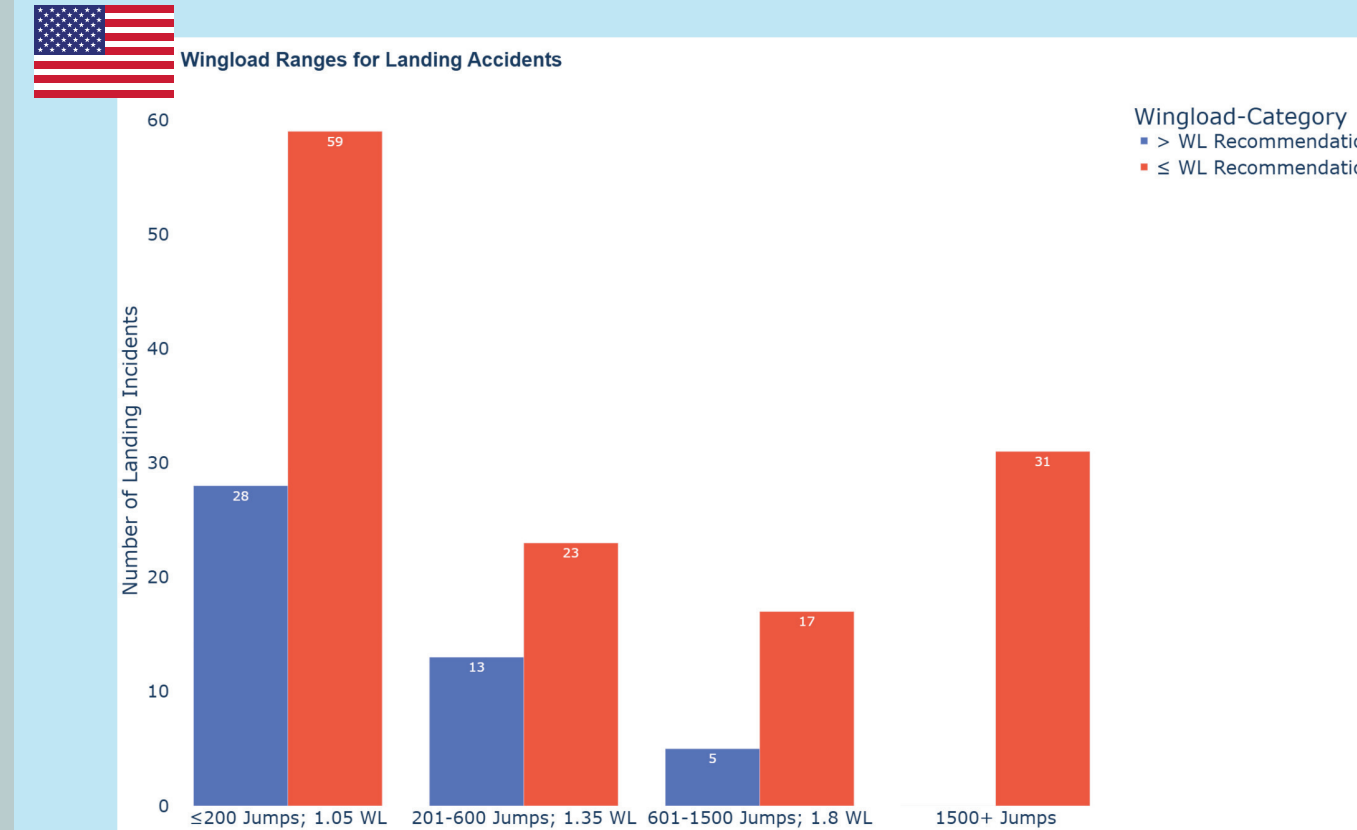
#### USA

- Very similar - inexperienced skydivers also face landing accidents the most.

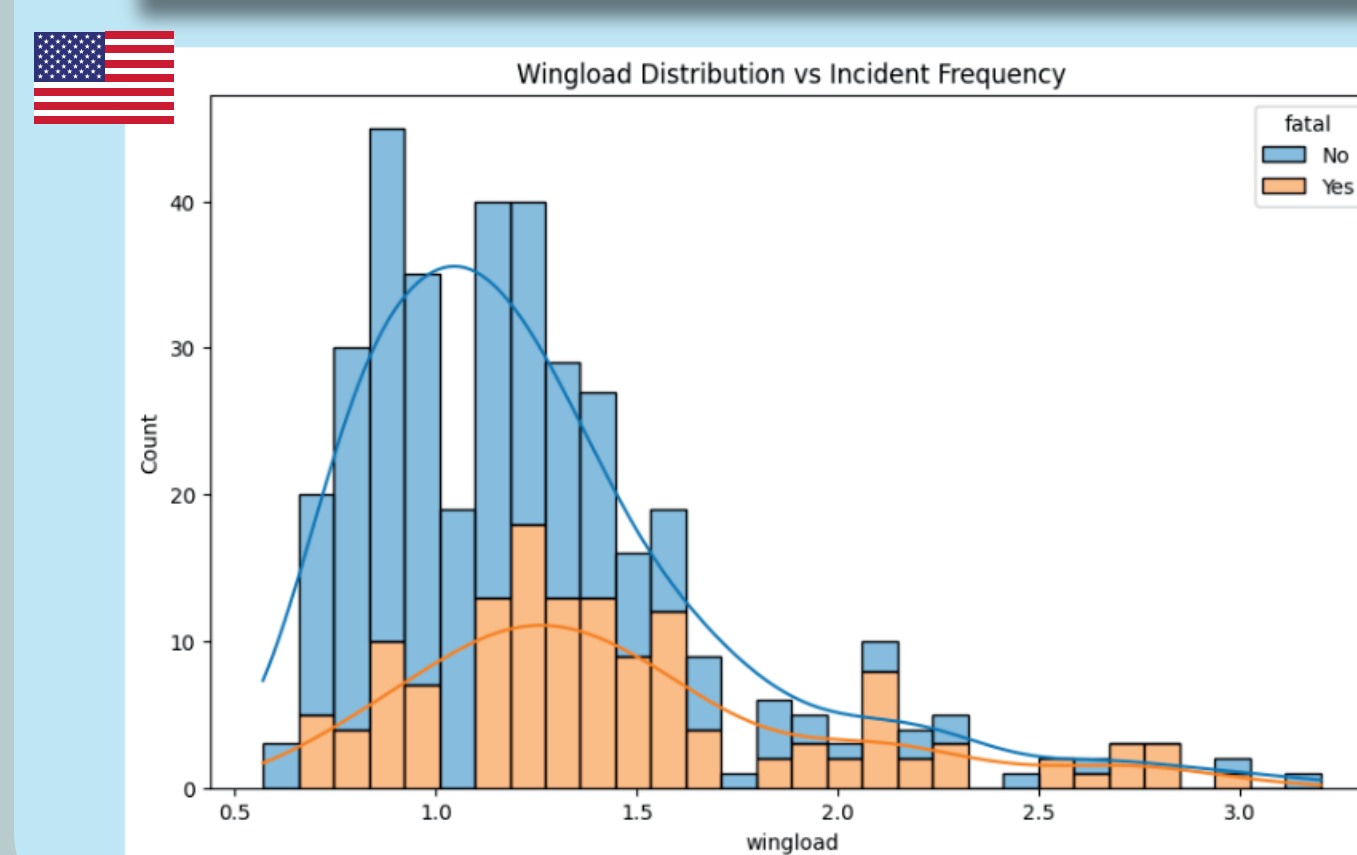


- Points towards a general problem, that should be addressed by better training.

### What role does Wingload play?

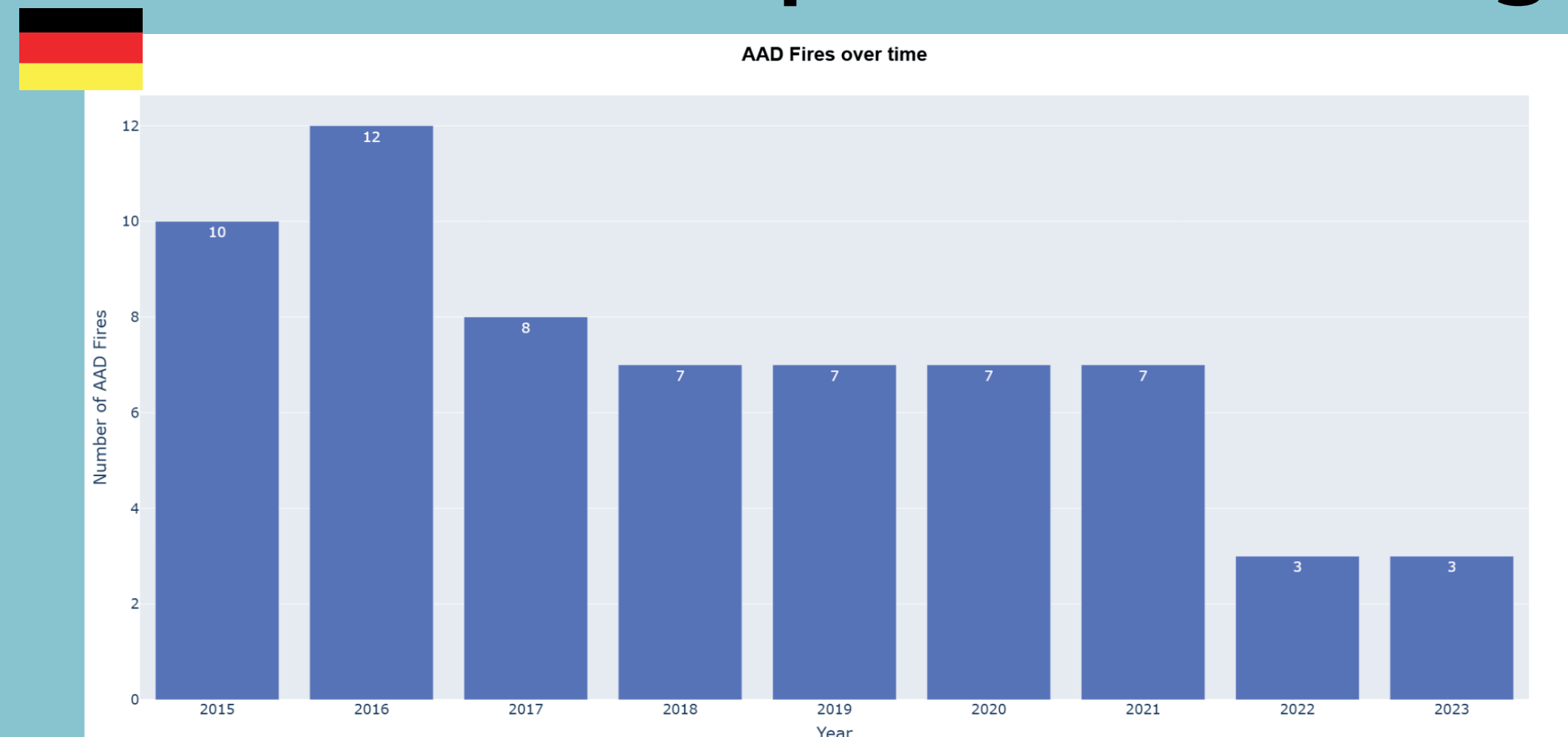


- In every jump range, more accidents happened within the recommended wingload range.
- Still, a third to half of the landing accidents happened with a higher than recommended wingload.
- This is likely a sign that overconfidence plays a major role in landing accidents.



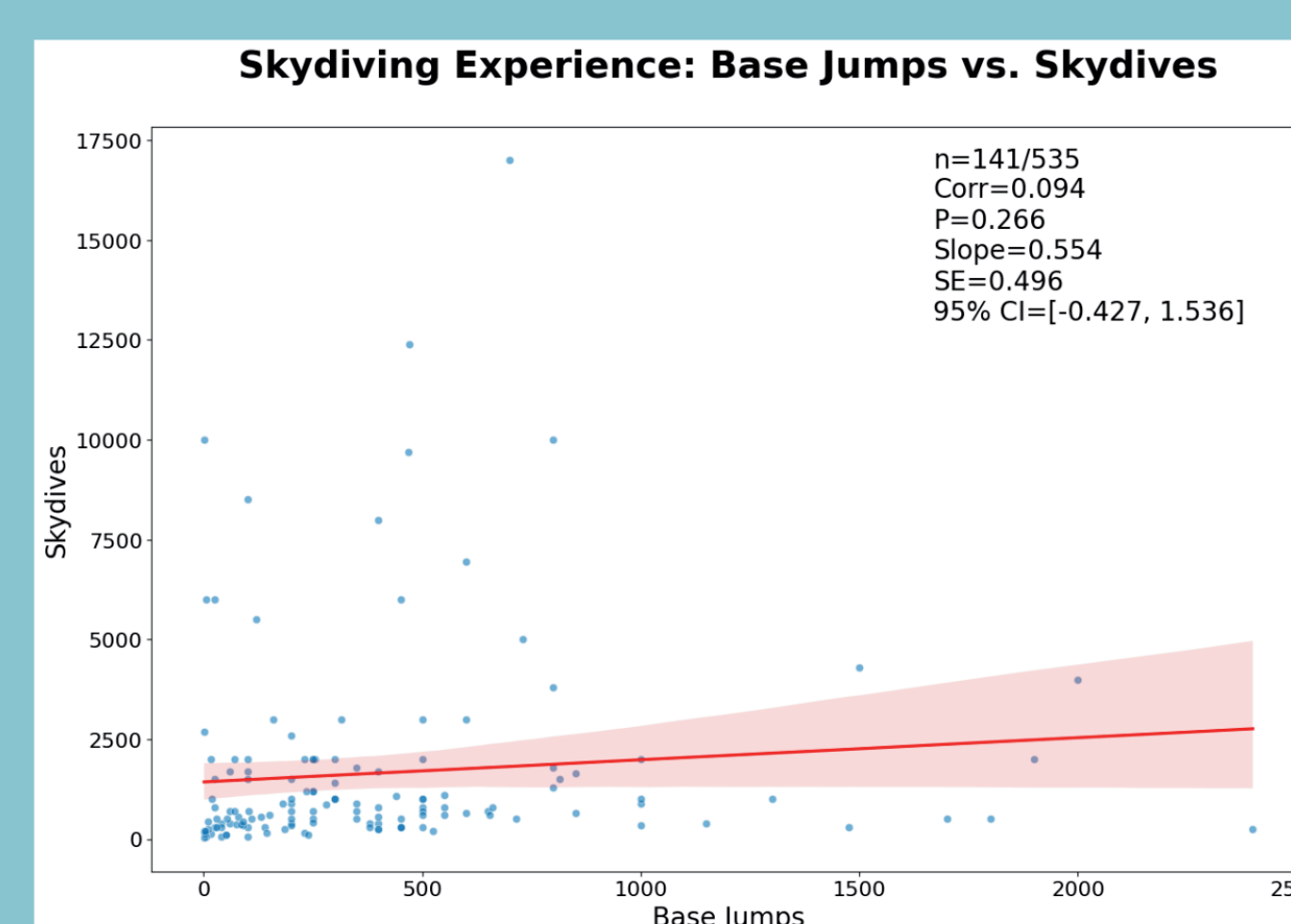
- Most incidents happen between wingloads of 0.8 and 1.4 (common for students and less experienced jumpers).
- The number of incidents drops with increasing wingload, but the share of fatal cases becomes more noticeable.
- Thus, higher wingloads may carry greater risk when something goes wrong.

### What is the impact of AAD usage?



- On average, 7.1 AAD uses occurred per year from 2015 to 2023 in Germany.
- Sometimes the activation was unintentional (fast low turns) or a manual activation occurred at the same time.
- Still, lives were saved in many of the cases, contributing to the often less than five fatal accidents a year in Germany.
- This indicates, that AADs being mandatory at most skydiving sites in Germany contributes to the safety of the sport.

### How does BASE jumping compare?



- Contrary to our assumptions, prior skydiving experience can't predict base jump survival rate in our analysis.
- Stark outliers in both directions (little skydives, many base jumps and vice versa) further point to this.
- **Correlation value:** 0.094 (minor positive correlation)
- **P-value** (statistical significance): 0.266 > 0.05 threshold ==> not significant

#### What factors lead to fatalities in BASE jumping?

- The causes for fatalities are manifold.
- Most common is proximity flying.



### Takeaways

- Almost all accidents can be attributed to human error - good training is key.
- Overall, accidents are very rare - difficult situations need to be trained.
- Technical aids like AAD save lives - they should be used.
- Like with any other activity - stay vigilant, check your gear and follow protocol.

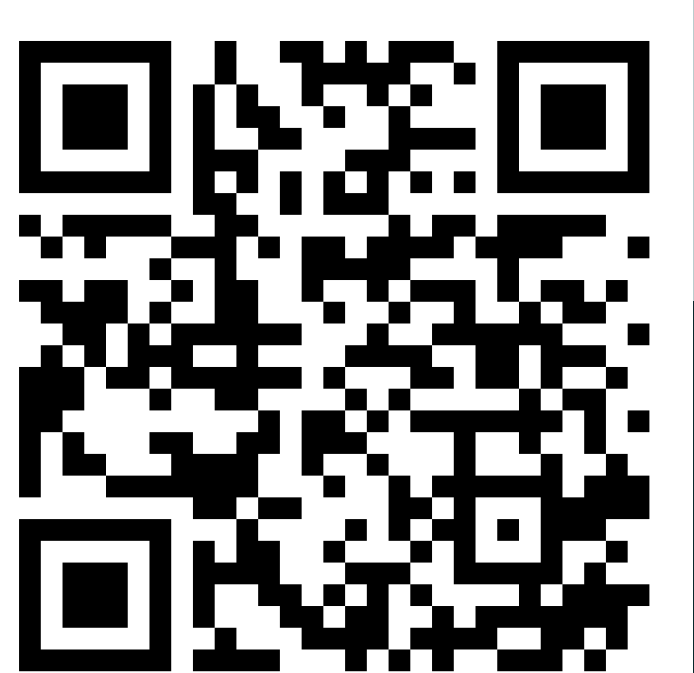


### What is ...

- ... an AAD? - Automatic activation device (deploys reserve chute).
- ... wingload? - The weight to canopy size ratio of the chute.
- ... BASE? - Jumping off high objects rather than a plane.

### Sources

- BFL: <https://bfl.baseaddict.com/list>
- USPA: <https://www.uspa.org/searchincidentreports>
- DFV: <https://www.dfv.aero/Info/Downloads/Auswahl/Tag/Unf%C3%A4lle>
- Statista: <https://www.statista.com/statistics/1088766>  
<https://www.statista.com/statistics/1084723>  
<https://de.statista.com/statistik/daten/studie/309004>  
<https://de.statista.com/statistik/daten/studie/308986>



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Code-Repository:  
<https://github.com/MH428/DSPProject>